

DRY AND WET WASTE SEGREGATOR

A Real-Time Project Report

Submitted to



Jawaharlal Nehru Technological University Hyderabad

In partial fulfillment of the requirements for the

award of the degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS & COMMUNICATION ENGINEERING

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INSTITUTE OF ENGINEERING AND TECHNOLOGY
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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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Certificate

This is to certify that the Real-Time Project Report on ***"DRY AND WET WASTE SEGREGATOR"*** submitted by **S.RAJU(24VE5A0407), M.JL.NARSIMHA (23VE1A0430), M.GANADEEP(23VE1A0431)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics & Communication Engineering** from Jawaharlal Nehru Technological University, Kukatpally, Hyderabad for the academic year 2024-25 is a record of bonafide work carried out by him / her under our guidance and Supervision.

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DECLARATION

We, **S. RAJU(24VE5A0407), M.JL. NARSIMHA(23VE1A0430), M. GANADEEP (23VE1A0431)** , hereby declare that the Real-Time Project titled "***DRY AND WET WASTE SEGREGATOR***" done by us under the guidance of **Dr.Ashok Battula**, which is submitted in the partial fulfillment of the requirement for the award of the B.Tech degree in **Electronics & Communication Engineering** at **Sreyas Institute of Engineering & Technology** for Jawaharlal Nehru Technological University, Hyderabad is our original work.

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ABSTRACT

This mini project aims to design and implement a low-cost, automatic Dry and Wet Waste Segregator suitable for household or small-scale use. With the growing concern over environmental sustainability, efficient waste management has become crucial. One of the key steps in waste management is the segregation of dry and wet waste at the source.

The system uses a moisture sensor to detect whether the waste is wet or dry. Once identified, the waste is automatically sorted into respective bins using a simple sorting mechanism, controlled by a microcontroller (such as Arduino). This automation minimizes human involvement, ensures better hygiene, and encourages the habit of waste segregation.

By implementing this system, households or industries can manage waste more effectively, thus contributing to environmental conservation, sustainable living and also aims to raise awareness about the importance of proper waste disposal. This project serves as a practical and scalable solution that can be integrated into smart waste management systems in the future.

This project highlights the potential of automation and smart systems in creating a cleaner and greener future.

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CHAPTER 1

INTRODUCTION

Waste management is a growing concern due to increasing population and urbanization. A key step in effective waste handling is the segregation of dry and wet waste. Dry waste includes recyclable materials like plastic, paper, and metal, while wet waste consists of biodegradable items like food and organic matter.

Manual segregation is often avoided due to health risks and inconvenience. To overcome this, our project proposes an automated Dry and Wet Waste Segregator using a moisture sensor and Arduino microcontroller. The system detects the type of waste based on moisture content and directs it into the appropriate bin using a servo motor.

This low-cost and eco-friendly solution reduces human effort, supports recycling and composting, and promotes better hygiene and environmental sustainability.

The project showcases how an Arduino Uno can be used to make decisions based on sensor input, making it ideal for beginners and enthusiasts looking to gain hands-on experience with embedded systems, basic electronics, and real-world automation.

1.1 About the Project

The *Dry and Wet Waste Segregator* is an innovative system designed to automate the process of waste classification at the source. It focuses on the efficient separation of household or industrial waste into two primary categories: dry waste (such as plastic, paper, metal, and glass) and wet waste (organic and biodegradable waste like food scraps and vegetable peels). Manual segregation is not only inefficient but also poses health risks to waste handlers.

The system uses sensors to detect the type of waste based on its properties, such as moisture content and material type. For instance, moisture sensors help identify wet (organic) waste, while metal detectors and infrared sensors can help classify dry (inorganic) waste like plastic, metal, and paper.

Once identified, the waste is directed to the appropriate bin using a mechanical setup like a servo-controlled flap or conveyor belt. This automation ensures faster and more hygienic waste segregation, reducing the burden on manual labor and promoting environmentally friendly disposal methods.

The implementation of this technology not only promotes environmental sustainability through better recycling and composting practices but also supports smart waste management solutions in both residential and urban settings.

1.2 Methodology

The methodology followed in the development of the *Dry and Wet Waste Segregator* involves designing and building an automated system capable of distinguishing between dry and wet waste. The process begins with identifying the core problem of inefficient and unhygienic manual waste segregation, which is common in most households and public spaces. To address this, a system was conceptualized that uses sensors and a microcontroller to automate the segregation process.

The system is built using key components such as a moisture sensor to detect the presence of water content in waste (indicating wet or organic waste). These sensors are connected to a microcontroller, such as an Arduino, which serves as the brain of the system. The microcontroller processes the sensor data and controls mechanical components like servo motors to direct the waste into appropriate bins.

After the initial setup, the system is tested with different types of waste to ensure accurate detection and sorting. Sensor thresholds are calibrated, and the control logic is fine-tuned to improve classification accuracy. Once all components are working reliably, the final prototype is assembled and demonstrated to show how mixed waste can be automatically separated into dry and wet categories.

1.3 Objectives

The primary objective of the project is to design and develop a system that can segregate the dry and wet waste segregator by using sensors and Arduino.

- To design and develop an automated system that can effectively segregate dry and wet waste at the source.
- To reduce human intervention in the waste segregation process and minimize health risks associated with manual waste handling.
- To promote eco-friendly practices by encouraging source-level waste management through a simple and accessible system.
- To build a cost-effective and scalable prototype that can be used in households, schools, or small communities for practical waste management.
- To increase public awareness about the importance of waste segregation and its role in sustainable environmental practices.

1.4 Existing System

In the current scenario, waste segregation is predominantly carried out manually, where individuals are responsible for separating dry and wet waste at the source, such as in households or public areas. This manual approach often results in improper segregation due to lack of awareness, negligence, or inconvenience. This results in mixed waste being collected, which is not only difficult to sort later but also reduces the effectiveness of recycling and composting processes. Dry waste like plastics, paper, and metals often gets contaminated with wet waste like food scraps and liquids, making recycling nearly impossible without extensive cleaning or processing.

Moreover, manual segregation puts sanitation workers at risk of exposure to harmful substances, infectious materials, and sharp objects, leading to serious health issues. The process is labor-intensive, time-consuming, and prone to errors. In addition, waste management systems in many areas still lack the infrastructure to support effective segregation practices.

CHAPTER 2

BLOCK DIAGRAM AND EXPLANATION

2.1 BLOCK DIAGRAM

The provided block diagram illustrates a system composed of several key components. At the center is an Arduino Uno, which serves as the primary controller for the entire setup. This microcontroller is connected to various peripherals and devices, each playing a crucial role in the system's functionality as shown in figure 2.1.

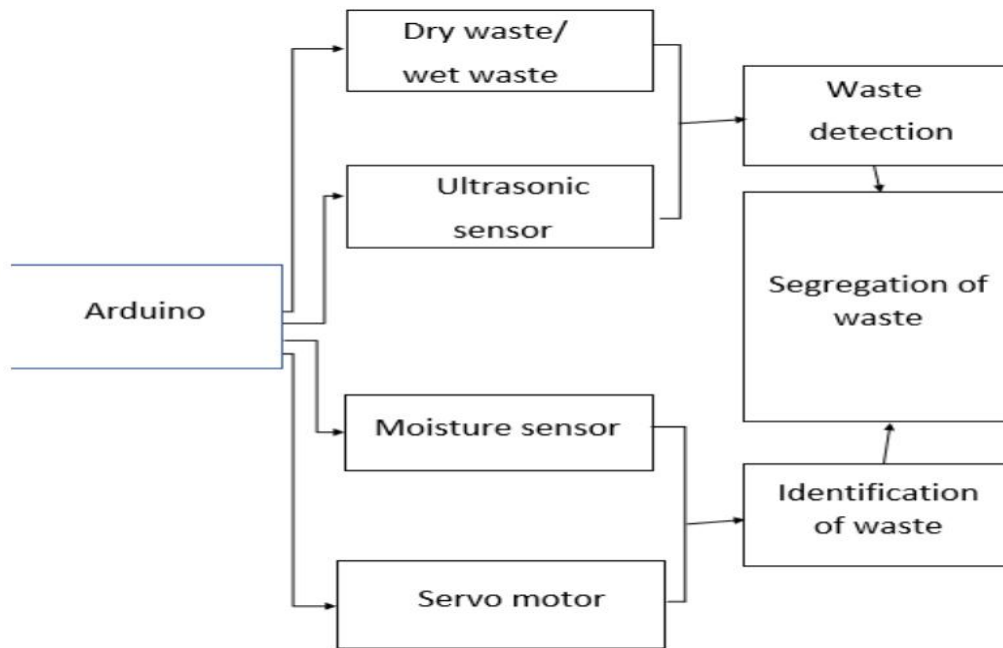


Figure. 2.1 Block Diagram

2.2 EXPLANATION

1. Input: Dry/Wet Waste

- ✓ The system starts when the user places any type of waste (dry or wet) into the waste inlet.
- ✓ This waste is passed through the sensor area for detection and identification.

2. Ultrasonic Sensor – Waste Detection

- ✓ This sensor checks whether any object (waste) is present in front of the sensor.
- ✓ It measures the distance of the waste from the sensor using ultrasonic waves.

- ✓ When the distance falls within a certain range, it signals the Arduino that waste has been inserted.
- ✓ It prevents unnecessary operation of the system when there is no waste.

3. Moisture Sensor – Identification of Waste

- ✓ After the waste is detected, it goes to the moisture sensor.
- ✓ This sensor checks for moisture content
- ✓ If moisture is high → it is identified as wet waste.
- ✓ If there is no or very low moisture → it is considered dry waste.
- ✓ The result is sent to the Arduino for decision-making.

4. Arduino – Main Controller

- ✓ The Arduino acts as the brain of the system.
- ✓ It receives:
 - ✓ Detection signal from the ultrasonic sensor.
 - ✓ Identification result from the moisture sensor.
- ✓ Based on this information, it sends commands to the servo motor to sort the waste properly.

5. Servo Motor – Waste Segregation

- ✓ The servo motor physically moves the waste into the correct bin:
- ✓ One bin for dry waste
- ✓ Another bin for wet waste
- ✓ The motor's movement is precisely controlled by the Arduino, depending on the moisture reading.

CHAPTER 3

SOFTWARE AND HARDWARE DESCRIPTION

3.1 SOFTWARE DESCRIPTION

The project involves the use of both hardware and software components to segregate the Waste into Dry and Wet. The hardware includes Arduino Uno, servo motor, Ultrasonic Sensor, Moisture Sensor and Battery. The software aspect revolves around the Arduino IDE.

3.1.1 Arduino IDE

The Arduino Uno Integrated Development Environment (IDE) is the software where we write and upload code to the Arduino Uno board, as shown in Fig. 3.1. It allows users to write programs in the Arduino programming language, which is based on C/C++.

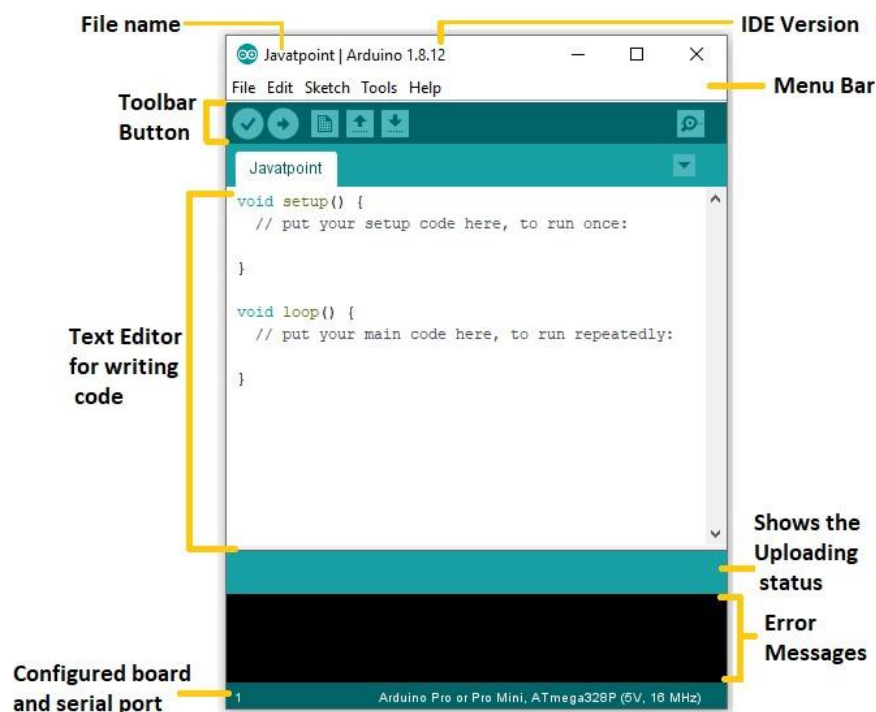


Figure. 3.1 Arduino Uno IDE

It includes a compiler to convert code into machine language and an uploader to transfer the program to the board via USB. A serial monitor allows real-time data exchange for debugging. The IDE is compatible with Windows, macOS, and Linux, making it accessible to a wide range of users.

The Arduino Uno IDE also facilitates the creation of sketches, which are the programs written for Arduino boards. It provides templates and examples to help beginners get started quickly. Users can access a vast collection of community contributed libraries, expanding the functionality of their projects. The IDE allows for easy management of board settings, enabling users to select different Arduino models. It supports version control through a built-in system, which helps track changes to sketches. The IDE can be customized with themes and preferences to enhance the user experience. Additionally, users can create their libraries for specific components. The IDE promotes a strong community, where users share projects and seek help. Overall, the Arduino Uno IDE is an essential tool for anyone interested in electronics and programming. Its simplicity and versatility make it suitable for both educational purposes and professional projects.

3.2 HARDWARE DESCRIPTION

3.2.1 Arduino Uno

The Arduino Uno is the heart of the Dry and Wet Waste Segregator. It is a micro controller board that reads the data from various sensors and processes it to control the actuators, like motors. The Arduino Uno is equipped with an ATmega328P micro controller and offers digital and analog pins to interface with other hardware components as shown in Fig.3.1.

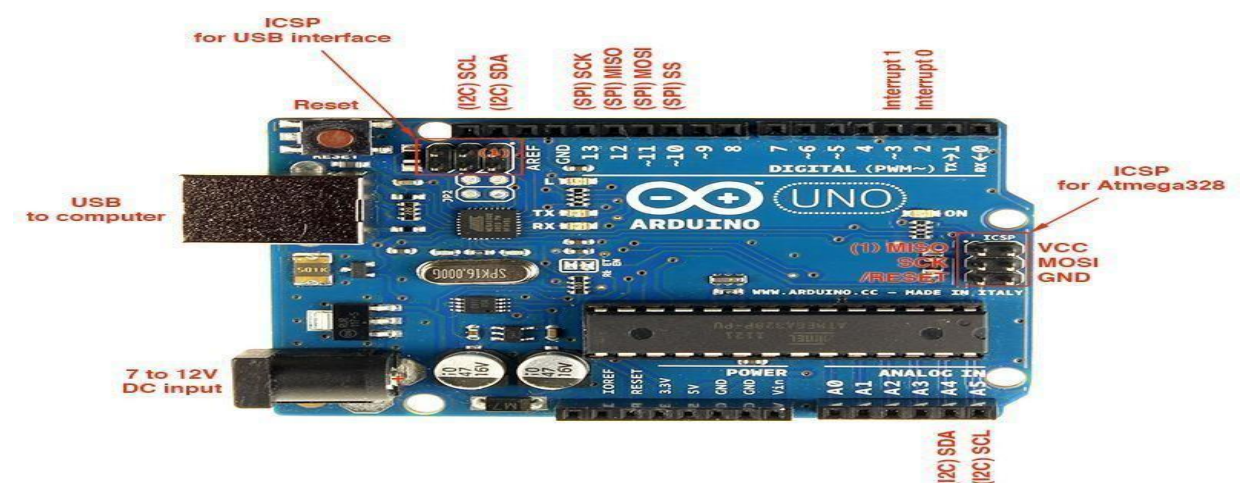


Figure. 3.2 Arduino Uno

The Arduino Uno serves as the central processing unit of the Light Chasing Robot machine, enabling it to function autonomously. It is built around the ATmega328P microcontroller, which provides a solid foundation for handling various tasks. With a variety of digital and analog pins, the Uno can interface with numerous sensors.

The board features 14 digital input/output pins, 6 analog inputs, and a USB connection for programming and power supply. This flexibility enables users to connect different components, including motors for adjusting speeds. The Arduino Uno operates at a voltage of 5V, making it compatible with a wide range of sensors and modules.

The compact size and low power consumption of the Arduino Uno make it suitable for Embedded applications, allowing it to be easily integrated into machines. Overall, the Arduino Uno's versatility, combined with its robust features, makes it an ideal choice for creating the Dry and Wet Waste Segregator that can adapt to various environments efficiently.

➤ **Specifications of Arduino Uno**

The Arduino UNO is a widely used open-source micro controller board. It serves as the brain of many electronics and robotics projects, offering simplicity, flexibility, and a large support community. The following are the specifications of Arduino UNO

- Microcontroller: ATmega328P
- Operating Voltage: 5V
- Input Voltage: 7- 12V (recommended)
- Digital I/O Pins: 14 (6 PWM outputs)
- Analog Input Pins: 6
- Flash Memory: 32 KB (of which 0.5 KB is used by the bootloader)
- USB Interface: Type B
- Dimensions: 68.6 mm x 53.4 mm

Programming the Arduino Uno is straightforward, thanks to the user-friendly Arduino IDE, where users can write and upload code to the board. The platform supports a variety of libraries that simplify the

process of integrating sensors and actuators, facilitating rapid development of complex algorithms. Additionally, the Arduino community provides extensive resources, tutorials, and forums, making it easy for users to troubleshoot issues and enhance their projects.

3.2.2 Servo motor(SG90)

A servo motor is a type of motor used for precise control of angular position, speed, and acceleration. It consists of a DC motor, gear system, control circuit, and a feedback mechanism (usually a potentiometer). In embedded projects like your waste segregator, servo motors are commonly used to move mechanical parts accurately, such as sorting flaps or rotating arms.

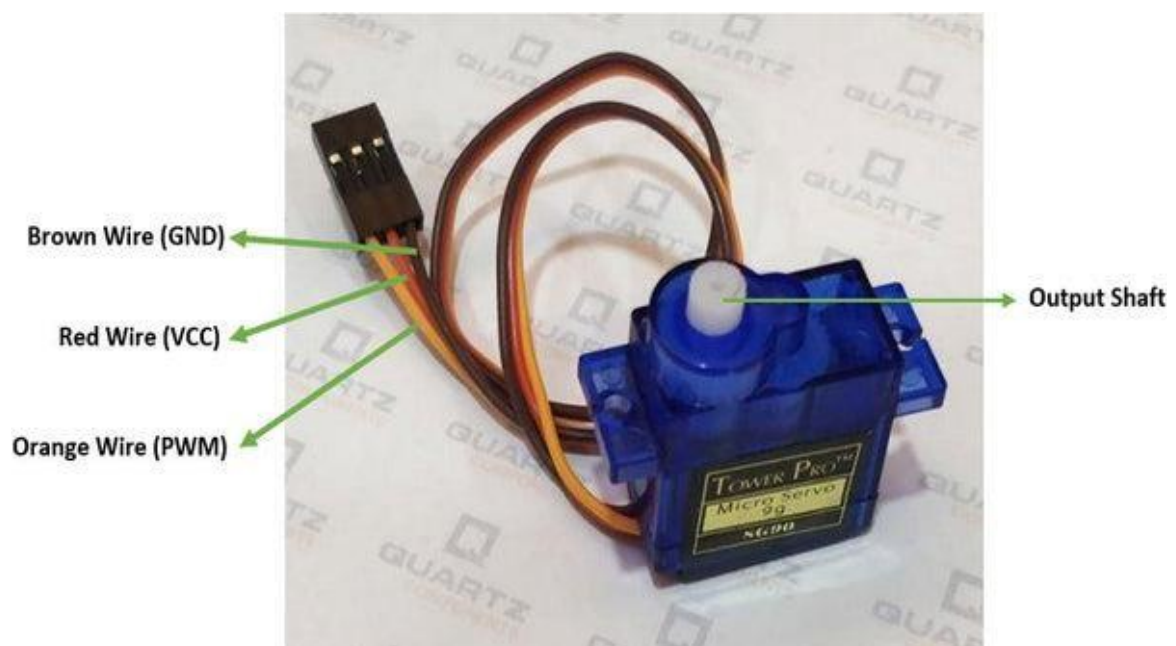


Figure. 3.3 Servo Motor

➤ **Pin Configuration of Servo Motor(SG90)**

The SG90 has a 3-pin connector, typically with the following color code:

Wire Color	Pin Name	Function
Orange	Signal	Receives PWM signal from microcontroller (e.g., Arduino) to control position.
Red	VCC/Power	Connects to +5V power supply.
Brown	GND	Connects to Ground (0V).

Table 3.1 Pin Configuration of Servo Motor

➤ **Specifications of Servo Motor**

Servo motor are small, low-cost DC geared motors widely used in embedded systems like waste segregators, commonly used to move mechanical parts accurately, such as rotating arms. They are ideal for light weight applications. The following are the specifications of the Servo Motor

- Model: SG90 Micro Servo
- Operating Voltage: 4.8V to 6.0V
- Stall Torque: ~1.8 kg-cm at 4.8V
- Operating Speed: ~0.1 sec/60° at 4.8V
- Rotation Angle: 0° to 180° (limited range)
- Control Signal: PWM(Pulse Width Modulation)
- Weight: Around 9 grams
- Gear Type: Plastic gears
- Motor Type: DC Motor (with control circuitry)
- Cable Length: ~25 cm (3-pin female connector)
- Dimensions: 22.2 mm x 11.8 mm x 31 mm
- Dead Band Width: ~10 μ s

3.2.3 5V Adapter

A 5V power supply adapter is an essential component used to provide a constant 5-volt DC output to power low-voltage electronic circuits and devices. It is widely used in embedded systems, Arduino-based projects, sensor modules, servo motors, and other low-power applications.

This adapter converts standard AC mains voltage (typically 100V–240V) into a regulated 5V DC output, which can safely power components that require stable voltage input. These adapters are available in various current ratings such as 1A, 2A, or 3A, depending on the power requirements of the connected devices.



Figure. 3.4 5V Adapter

➤ Pin Configuration

A typical 5V adapter with a barrel connector has two terminals:

Part	Polarity	Function
Inner Pin (Tip)	Positive(+)	Supplies +5V DC
Outer Sleeve	Negative (-)	Ground (GND)

Table 3.2 Pin Configuration of 5V Adapter

This is called "Center Positive" configuration, which is the most commonly used standard for 5V adapters.

➤ **Specifications of 5V Adapter**

A 5V Adaptor is a commonly used power source in electronics, and embedded systems. It provides a stable and sufficient voltage to drive DC motors, motor driver modules (such as Servo motor), sensors, and micro controllers like the Arduino when regulated properly. The following are the specifications of 5V Adapter

- Input Voltage: 100V-240V AC (standard household AC)
- Output Voltage: 5V DC (regulated)
- Current Output: Typically 1A, 2A or 3A (depending on model)
- Output Power: 5W to 15W
- Connector Type: Barrel Jack (5.5mm outer, 2.1mm inner)/ USB Type-A
- Polarity: Center Positive (+), Outer Sleeve Negative (-)
- Efficiency: Typically > 75%
- Ripple & Noise: <100 mVp-p (for regulated adapters)
- Protection Features: Over voltage, Over current, Short Circuit Protection
- Cable Length: Typically 1 meter
- Enclosure: Plastic casing, compact and insulated
- Cooling Method: Natural air cooling (no fan)

3.2.4 Ultrasonic sensor

An ultrasonic sensor is an electronic device that measures the distance to an object using sound waves. It emits a high-frequency ultrasonic pulse (typically 40 kHz), waits for it to bounce back from a surface, and then calculates the distance based on the time delay.

The ultrasonic sensor is a non-contact distance measuring device. It operates on the principle of echolocation, similar to how bats and dolphins navigate.

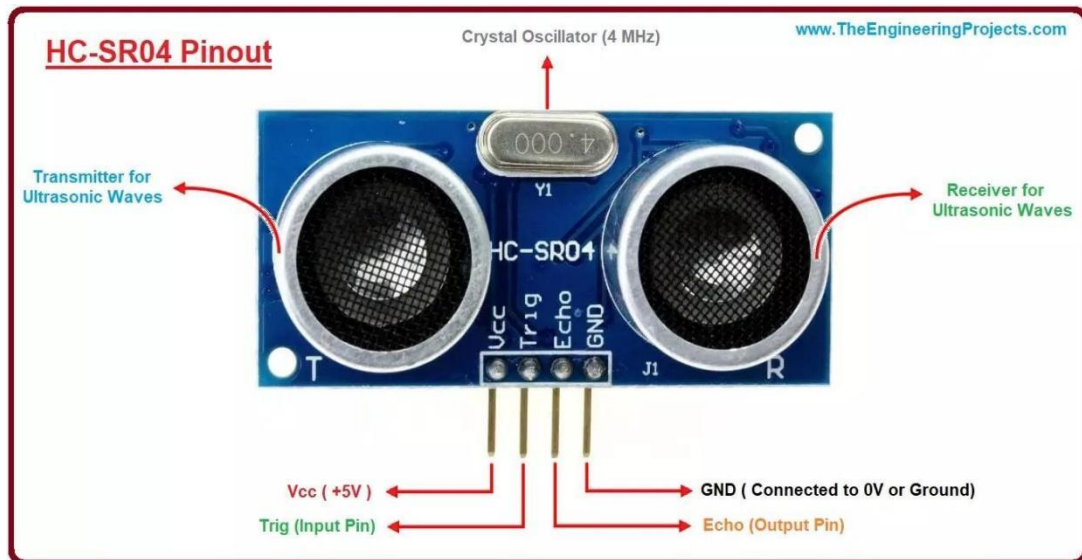


Figure 3.5 Ultrasonic Sensor

➤ Pin configuration

Pin	Name	Function
1	VCC	Power supply – Connect to +5V on Arduino or external supply
2	Trig	Trigger pin – Sends out an ultrasonic pulse
3	Echo	Echo pin – Receives the reflected signal
4	GND	Ground – Connect to GND of Arduino or circuit

Table 3.3 Pin Configuration of Ultrasonic Sensor

➤ Specifications of Ultrasonic sensor

Ultrasonic sensor is easy to integrate with microcontrollers like Arduino. Ultrasonic sensor has reliable distance sensing in non-contact applications. It is not affected by color or transparency of objects (unlike IR sensors).

- Model: HC-SR04
- Operating Voltage: 5V DC
- Operating Current: 15mA (typically)
- Operating Frequency: 40 kHz
- Measuring Range: 2 cm to 400 cm (0.02 m to 4.00 m)
- Accuracy: ± 3 mm

- Resolution: ~1 mm
- Measuring Angle: Approx. 15°
- Trigger Input Signal: 10 μ s TTL pulse
- Echo Output Signal: TTL signal; pulse width corresponds to distance
- Response Time: Less than 50 ms
- Dimensions: 45 mm x 20 mm x 15 mm (L x W x H)
- Interface Type: 4-pin (VCC, Trig, Echo, GND)
- Temperature Range: 0°C to 70°C

3.2.5 Moisture Sensor

A moisture sensor is an electronic device used to detect the water content in soil or other materials. It typically consists of two conductive probes that measure the resistance or conductivity of the material between them.

When moisture is present, conductivity increases and resistance decreases, allowing the sensor to determine how wet or dry the environment is. In the context of a waste segregation project, a moisture sensor helps identify wet waste (which has high moisture content) from dry waste, enabling accurate and automated sorting.

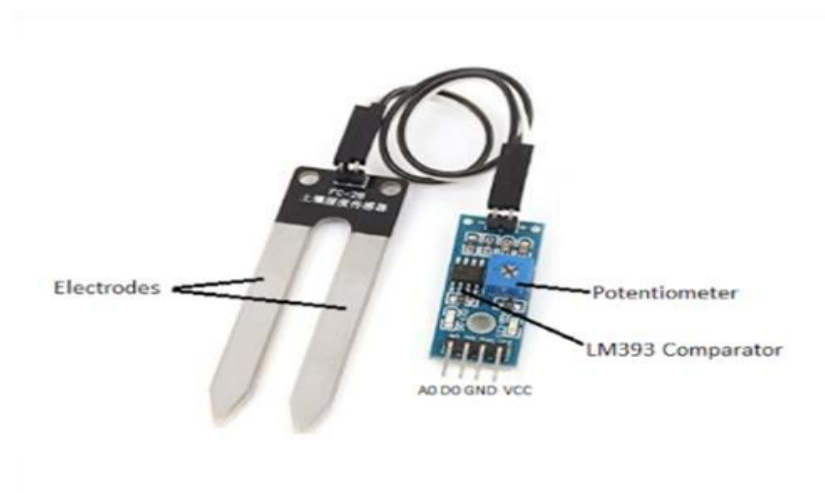


Figure. 3.6 Moisture Sensor

3.5.1 Pin Configuration

The pin configuration of a typical Soil Moisture Sensor module (often used in Arduino projects), which includes both the sensor probe and the interface module:

Pin	Name	Function
VCC	Power	Connect to 3.3V or 5V (typically 5V)
GND	Ground	Connect to GND of microcontroller
AO	Analog Out	Outputs an analog voltage based on moisture level
DO	Digital Out	Digital Out (only in 4-pin versions; for threshold-based output)

Table 3.4 Pin Configuration of Moisture Sensor

➤ Specifications of Moisture Sensor

Moisture sensor is simple and low-cost moisture detection. Moisture Sensor is suitable for indoor use and educational projects. It is easy to integrate with Arduino. Moisture Sensor helps distinguish wet waste from dry waste in segregation systems. Specifications of a typical Soil Moisture Sensor (with analog output module), commonly used in Arduino

- Operating Voltage: 3.3V to 5V DC
- Operating Current: < 20 mA
- Output Type: Analog (AO) and Digital (DO – if 4-pin version)
- Sensor Type: Resistive
- Measurement Principle: Measures conductivity between two probes
- Analog Output Range: 0 – 1023 (10-bit ADC values on Arduino)
- Digital Output: High or Low (based on set threshold, if DO pin available)
- Adjustable Sensitivity: Yes (via potentiometer on module – only for Do pin)
- Probe Material: Typically nickel-plated or copper alloy
- Dimensions: Probe: ~60 mm x 20 mm; Module: ~30 mm x 16 mm
- Working Temperature: 10°C – 30°C (room temperature use)
- Cable Length: Usually ~20 cm (if needed)

CHAPTER 4

WORKING AND RESULTS

4.1 Working:

The Dry and Wet Waste Segregator is an automated system designed to classify waste based on its moisture content. The process begins when a user places a piece of waste into the input section of the bin. An ultrasonic sensor detects the presence of the object and triggers the system. Once detected, a moisture sensor comes into contact with the waste to measure its moisture level. Based on the sensor reading, the system identifies whether the waste is wet or dry.

The classification result is then passed to a microcontroller (such as an Arduino), which controls a servo motor connected to a flap or divider. The motor rotates to direct the waste into the correct compartment — either the dry bin or the wet bin. The system resets automatically, ready to sort the next item. You can see the circuite connections as shown in figure 4.1.

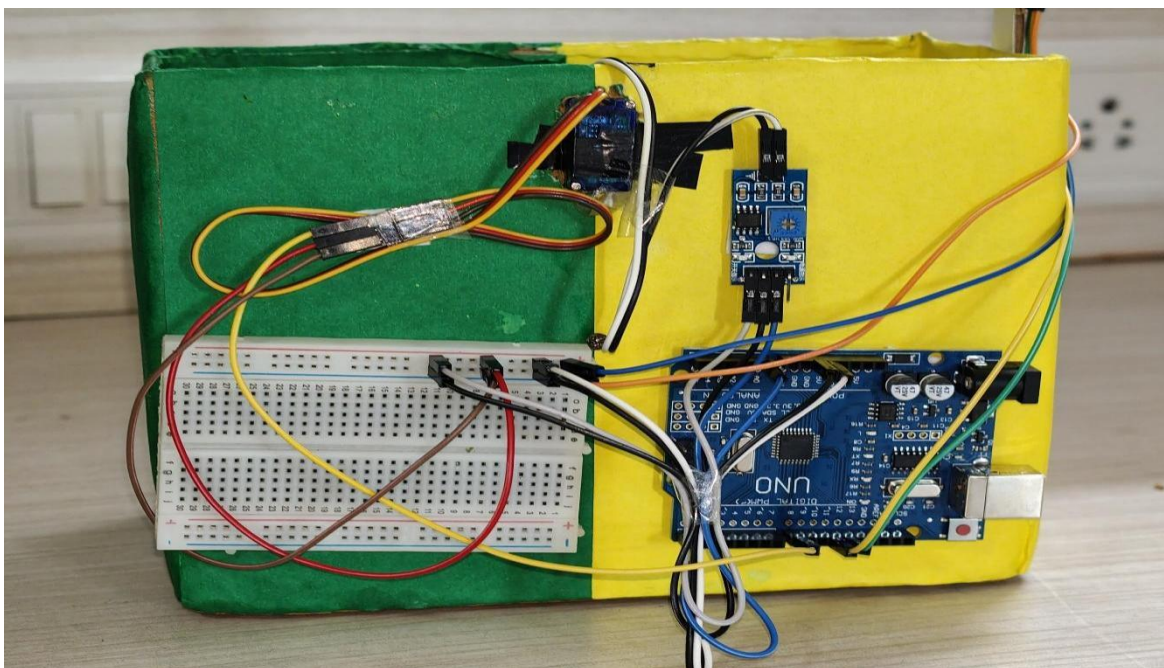


Figure.4.1 Circuite Connections

The provided circuit diagram illustrates a system composed of several key components. At the center is an Arduino Uno, which serves as the primary controller for the entire setup. This microcontroller is connected to various peripherals and devices, each playing a crucial role in the system's functionality.

- ✧ **Power Supply:** The 5V adapter provides a stable DC power supply to all the electronic components in the system. It is connected to the Arduino board and supplies sufficient voltage to operate the ultrasonic sensor, moisture sensor, and servo motor. A consistent power source is essential for the reliable functioning of the segregator.
- ✧ **Ultrasonic Sensor:** The ultrasonic sensor is used to detect the presence of waste when it is placed inside the bin. It emits ultrasonic waves and measures the time taken for the echo to return. If an object is detected within a specific range, the sensor sends a signal to the Arduino to start the sorting process. This sensor ensures that the system only activates when waste is present.
- ✧ **Microcontroller (Arduino Uno):** The Arduino Uno acts as the brain of the system. It receives inputs from the ultrasonic and moisture sensors, processes the data, and makes decisions accordingly. Based on the moisture content of the waste, it sends appropriate signals to the servo motor to direct the waste to the correct bin. The Arduino executes all logical operations and controls the entire workflow.
- ✧ **Moisture Sensor:** The moisture sensor is a key component that detects the water content in the waste material. It has two conducting probes that measure the electrical conductivity of the waste. Wet waste conducts electricity better than dry waste, allowing the sensor to differentiate between the two. The sensor sends an analog signal to the Arduino based on the moisture level detected.
- ✧ **Servo Motor:** The servo motor is responsible for physically moving the flap that directs the waste into the correct bin. It receives angle-based rotation commands from the Arduino. If the waste is dry, the servo moves the flap in one direction; if it is wet, it rotates to guide

the waste in the opposite direction. It ensures precise and reliable mechanical action in the system.

➤ **Powering the System**

The entire Dry and Wet Waste Segregator system is powered using a 5V DC power adapter. This adapter provides a stable voltage supply necessary for the operation of all components, including the Arduino Uno, ultrasonic sensor, moisture sensor, and the servo motor. The 5V output is ideal for microcontroller-based projects and ensures safe operation without overheating or overloading any components.

The adapter is typically connected through a DC barrel jack or the VIN and GND pins on the Arduino board. Proper grounding is maintained across all components to ensure consistent performance. This simple yet efficient power setup ensures the system remains reliable and autonomous during continuous operation.

4.2 RESULTS:

The results of the Dry and Wet Waste Segregator can be evaluated based on the system's performance in automating various tasks, while ensuring smooth and efficient operation. Here's a detailed breakdown of the results:

➤ **Wet Box**

When the Moisture Sensor detect density of item, the system dump the item according to the density. If the density is high the system will dump item into Wet box is shown in Fig. 5.2.



Figure.4.2 Wet Box

➤ **Dry Box**

When the Moisture Sensor detects that the density of the item is less then the system will dump the item into Dry box is shown in Fig. 5.3.



Figure.4.3 Dry Box

ADVANTAGES AND DISADVANTAGES

Advantages :

The Dry and Wet Waste Segregator offers several practical and environmental benefits that enhance the technology experience. The following are some key advantages of using the Dry and Wet Waste Segregator:

- ✧ Automatic waste segregation reduces the need for manual sorting.
- ✧ Minimizes health risks for sanitation workers by avoiding direct contact with waste.
- ✧ Promotes eco-friendly waste management and recycling practices.
- ✧ Simple and cost-effective design using easily available electronic components.
- ✧ Low maintenance and easy to operate for users of all ages. Educational value — ideal for teaching automation and sensor-based systems.

Disadvantages :

Dry and Wet Waste Segregator, like any other technology, has its limitations and drawbacks. While they offer numerous benefits, it's essential to consider the potential disadvantages before making a purchase.

- ✧ Outdoor use is limited, as sensors and electronics may not be weatherproof without added protection.
- ✧ Depends on a constant power supply — system won't work during power outages without backup.
- ✧ Mechanical parts (like servo motor) may wear out over time, requiring maintenance.
- ✧ Scaling the system for industrial or large-scale use may require more advanced sensors and design changes.

CHAPTER 5

APPLICATIONS

1. **Light-chasing robots** are autonomous systems that detect and move toward light sources, and they can be utilized in various practical and educational scenarios, including:
2. **Household Waste Management:** This project can be installed in homes to automatically separate dry and wet waste from daily household garbage. It reduces the burden of manual sorting and ensures that biodegradable and non-biodegradable waste is disposed of properly, making waste disposal easier and eco-friendly for families
3. **Educational Institutions:** Schools and colleges can use this project for two main purposes — to manage waste efficiently on campus and as a learning tool. It helps raise awareness about the importance of waste segregation among students while also being a great hands-on project for teaching electronics, sensors, and automation.
4. **Offices and Workplaces:** In commercial buildings and offices, waste segregators can help ensure that waste is sorted correctly at the source. This supports corporate environmental responsibility and contributes to greener workspaces by encouraging proper disposal habits among employees.
5. **Smart City Initiatives:** This system fits well into smart city frameworks, where automation and sustainability are key goals. It can be integrated into public smart bins to reduce manual effort, improve cleanliness in public areas, and assist municipal authorities in managing waste more effectively.
6. **Hospitals and Clinics:** While this system may not be suitable for hazardous or biomedical waste, it can still be used in areas where general, non-hazardous waste is generated. This helps in maintaining a clean environment and improving hygiene standards.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

In conclusion, the Dry and Wet Waste Segregator project presents a smart, efficient, and eco-friendly approach to handling household and public waste by automating the segregation process. Using sensors like the ultrasonic sensor to detect object presence and a moisture sensor to distinguish between wet and dry waste, the system reduces the dependency on human intervention and encourages cleaner surroundings.

The integration of a servo motor allows for smooth mechanical control, directing the waste into the appropriate bin with precision. This project not only simplifies waste management but also promotes better hygiene, supports recycling efforts, and reduces environmental pollution. It is cost-effective, energy-efficient, and suitable for a wide range of applications — from homes and schools to offices and smart cities.

Looking ahead, the project holds significant future potential for innovation and expansion. It can be improved by incorporating machine learning and image processing to identify different materials such as plastics, glass, metals, and biodegradables. Adding multiple sensors like gas sensors (for detecting harmful waste), weight sensors, and color sensors can enhance classification accuracy.

IoT integration would allow users or municipal workers to remotely monitor bin status in real-time through mobile apps or dashboards. For large-scale use, the model can be upgraded into industrial versions for use in recycling plants or smart bins in public places. Additionally, the waste segregator can be integrated with smart composting systems for wet waste or shredding mechanisms for dry waste, turning it into a multi-functional waste processing unit.

Overall, the Dry and Wet Waste Segregator has the potential to continuous development, this system can become a vital component in achieving clean, smart, and sustainable urban living and can help in many sectors.

REFERENCES

- 1.**Arduino Official Website** –(For technical specifications and programming of Arduino Uno).
- 2.**SG90 Servo Motor** -(For understanding pin configuration and specifications).
- 3.**HC-SR04 Ultrasonic Sensor Guide** –
(For working principle, circuit diagram, and coding reference).
- 4.**Soil Moisture Sensor (FC-28) Datasheet and Guide** – (Used as a moisture detector for waste classification).
- 5.**Waste Management Rules – Government of India** – (To understand government policies and need for waste segregation).
- 6.**Smart Waste Management Concepts – Research Papers & IEEE Xplore** (For reference on how this project fits into smart city and automation themes).

APPENDIX

```
#include <Servo.h>
Servo servo1;
const int trigPin = 12;
const int echoPin = 11;
long duration;
int distance=0;
int potPin = A0;
int soil=0;
int fsoil;
int maxDryValue=1;      //this value decides how much
                          humidity require to treat waste as wet object
int Ultra_Distance=15;  //set Distance between
                          Ultrasonic and moisture(soil) sensor in cm
void setup()
{
    Serial.begin(9600);
    pinMode(trigPin, OUTPUT);
    pinMode(echoPin, INPUT);
    servo1.attach(8);
    Serial.println("Soil Sensor Ultrasonic Servo");
}
void loop()
{
    int soil=0;
    for(int i=0;i<2;i++)
    {
        digitalWrite(trigPin, LOW);
        delayMicroseconds(7);
        digitalWrite(trigPin, HIGH);
        delayMicroseconds(10);
        digitalWrite(trigPin, LOW);
        delayMicroseconds(10);
        duration = pulseIn(echoPin, HIGH);
        distance= duration*0.034/2+distance;
        delay(10);

    }
    distance=distance/2;

    if (distance <Ultra_Distance && distance>1)
    {
        delay(1000);
        for(int i=0;i<3;i++)
        {
```



```

soil = analogRead(potPin) ;
soil = constrain(soil, 485, 1023);
fsoil = (map(soil, 485, 1023, 100, 0))+fsoil;
delay(75);
}
fsoil=fsoil/3;

Serial.print("Humidity: ");
Serial.print(fsoil);
Serial.print("%");
Serial.print("  Distance:");
Serial.print(distance);Serial.print(" cm");
if(fsoil>maxDryValue)
{delay(1000);
  Serial.println("    ==>WET Waste ");
  servo1.write(170);
delay(3000);}
else{ delay(1000);
  Serial.println("    ==>Dry Waste ");
servo1.write(10);
delay(3000);}

servo1.write(90);}
distance=0;
fsoil=0;delay(1000);
}

```